

Machine Learning: Unit 1 Linear Models and Regression

VTU 2022 Scheme CBCS - BCS602 Module 1

--- MODULE OVERVIEW & LEARNING OBJECTIVES ---

1. Introduction to Machine Learning: Supervised vs Unsupervised vs Reinforcement Learning
2. Linear Regression: Hypothesis function, Cost Function [MSE], and Normal Equation
3. Batch Gradient Descent vs Stochastic Gradient Descent [SGD]
4. Logistic Regression for Binary Classification: Sigmoid activation, Log-Loss
5. Regularization Techniques: Ridge Regression [L2] and Lasso Regression [L1]

--- IMPORTANT VTU 10-MARK QUESTIONS ---

- Q1. Derive the Gradient Descent weight update equation for Linear Regression [10 Marks]
- Q2. Explain Logistic Regression decision boundary with solved example [10 Marks]